

Quantum Communication Can Be Exponentially Weaker than Public Coins in Multiparty SMP

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Abstract

The Simultaneous Message Passing (SMP) model provides a fundamental framework for studying the relative power of classical and quantum communication in theoretical computer science. In the two-player setting, it is well-known that shared randomness (the public-coin model) and quantum communication are incomparable. For instance, Gavinsky et al. (STOC 2006) exhibited a relational problem, INDEX COORDINATION, that can be solved by a public-coin SMP protocol using $O(\log n)$ bits, whereas any quantum SMP protocol solving it requires $\Omega(n^{1/2})$ qubits in the unambiguous model, where the referee may output a special failure symbol but never a wrong answer.

In this work, we establish an exponential separation between the public-coin and quantum SMP models for a natural k -party generalization of INDEX COORDINATION, which we denote $IC_{k,n}$. We show that a public-coin protocol can solve the k -party relation $IC_{k,n}$ using $O(k \log n)$ bits, whereas any unambiguous quantum protocol without shared entanglement requires $\Omega(\frac{1}{k} n^{1-1/k})$ qubits. This yields an exponential separation for constant $k = O(1)$, and a polynomial separation for $k = \Theta(\log n)$.

Conceptually, our result shows that, in the multiparty SMP model, the coordination afforded by public randomness cannot, in general, be efficiently simulated by quantum superposition. Combined with prior separations in the opposite direction, where multiparty quantum protocols outperform public-coin protocols, this establishes that the exponential incomparability between the types of resources extends in a non-trivial way to arbitrary values of k , scaling with the number of players, beyond the previously studied case $k = 2$.

The lower bound for the two-party case relies on a straightforward bound for the referee's success probability. Nevertheless, in the multiparty setting, we require a much more involved argument for bounding the success probability of measurements applied to product states. To this end, we prove a theorem on unambiguous quantum state identification, which may be of independent interest for the study of quantum lower bounds in simultaneous communication.

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1 Introduction

The Simultaneous Message Passing (SMP) model, introduced by Yao [25], is a standard framework for analyzing resource trade-offs in distributed computing. In this setting, k players receive a private input $x_j \in \{0, 1\}^n$, where $j \in [k]$, and each of them sends a single message m_j to a referee. The referee, who possesses no input, must solve a computational problem, defined on $(x_j)_{j \in [k]}$, based solely on the messages $(m_j)_{j \in [k]}$. The cost of the protocol is defined as the length of the longest message sent by any single player.

A central objective in this area is to determine the comparative power depending on three distinct types of resources: classical private randomness, classical public randomness, and quantum superposition.

The hierarchy of communication resources is well-established in the two-party setting. Classically, public randomness is strictly more powerful than private randomness. This separation is best exemplified by the EQUALITY function, where public coins enable a constant communication cost of $O(1)$ bits, whereas protocols restricted to private randomness require $\Omega(n^{1/2})$ bits [1, 17]. In the quantum domain, protocols can simulate private randomness by measuring states in superposition; consequently, quantum communication is at least as powerful as the private-coin model. Moreover, it can be strictly more powerful: Buhrman et al. [5] demonstrated that quantum messages can solve EQUALITY with logarithmic cost, achieving an exponential advantage over classical private randomness via quantum fingerprinting. Therefore, a central problem in this area is to determine whether quantum communication without shared entanglement can simulate the coordination capabilities of *public* randomness.

For the two-party case, Gavinsky et al. [11] answered this in the negative for the zero-error regime (also known as unambiguous protocols), where the referee either outputs a valid solution with probability at least $2/3$, or returns the inconclusive symbol $\perp$, and never outputs a wrong answer. They introduce the relational problem INDEX COORDINATION with cost $O(\log n)$ in the classical shared randomness model while showing a lower bound of $\Omega(n^{1/2})$ for quantum communication. This separation was extended in [12] by proving a quantum lower bound of $\Omega(n^{1/3})$ for the same problem but in a less restrictive version of the model where the referee is able to output a wrong answer with small probability. This result is complementary to the separation shown in [4], where the HIDDEN MATCHING relational problem was introduced, allowing to prove a separation where quantum protocols can achieve $O(\log n)$ bits, while public coins are proved to need $\Omega(n^{1/2})$ bits. Together, these exponential separations show that, within the two-party setting, the quantum and shared-randomness models are incomparable.

Extending such separations to the multiparty setting requires careful consideration of the role of the number of players, k . While it is trivial to extend two-party separations by simply adding $k - 2$ dummy players, such constructions offer little insight into the nature of multiparty distributed computing. The true challenge lies in establishing separations for multiparty problems where the communication complexity depends non-trivially on k , and where the gap between resources intensifies as more players are added. However, proving such scalable lower bounds is difficult; techniques like discrepancy or multidimensional norms [18] often weaken or become intractable when applied to an arbitrary number of parties.

While there are no known separations for total boolean functions, there exist separations on relational problems for the general bounded-error model, where referee output is allowed to be wrong. The *zero-error* regime serves as an entry point for establishing an exponential separation where quantum protocols are weaker in the multiparty setting.

In this work, we establish an exponential separation between the public-coin and quantum SMP models for a k -party generalization of INDEX COORDINATION, that we denote $IC_{k,n}$. We show that a public-coin protocol can solve the k -party relation $IC_{k,n}$ using $O(k \log n)$ bits, whereas any unambiguous quantum protocol without shared entanglement requires $\Omega(\frac{1}{k} n^{1-1/k})$ qubits. This yields an exponential separation for constant $k = O(1)$, and a polynomial separation for $k = \Theta(\log n)$. Conceptually, our result shows that, in the multiparty SMP model, the coordination afforded by public randomness cannot, in general, be efficiently simulated by quantum superposition. This stands in contrast to the previously known generalization of the HIDDEN MATCHING problem to the k -party setting [13], which demonstrates the converse separation where quantum protocols are exponentially more efficient than classical ones. Combined with this prior result, our work establishes that the incomparability between the types of resources is a phenomenon that scales with the number of players.

1.1 Our Contribution

In this work, we establish an exponential separation between the public-coin classical model and the zero-error quantum model in the multiparty SMP setting. To achieve our goal, we extend the separation originally presented by Gavinsky et al. in [11, 12] for two players to an arbitrary number of players. To do this, we extend their combinatorial construction by developing an information-theoretic tool.

Exponential Separation in Multiparty SMP

We introduce the multiparty problem INDEX COORDINATION ($IC_{k,n}$), defined as follows: the first $k - 1$ players receive a string $x_j \in \{0, 1\}^n$. The k -th player receives a string $x_k \in \{0, 1\}^n$ together with a selector string $s \in \{0, 1\}^n$ with exactly n/k positions set to 1. The referee must output a tuple $(i, x_1^{(i)}, \dots, x_k^{(i)})$ for a valid index i such that $s^{(i)} = 1$. For the particular case when $k = 2$, $IC_{2,n}$ corresponds to the original relation introduced in [11].

Intuitively, the difficulty of this problem lies in the players' inability to coordinate on the target index i without shared randomness. In the public-coin model, players can use shared random bits to agree on a small sequence of candidate indices to check; since the selector string has a relative weight of $1/k$, they find a valid intersection with high probability using very little communication. In contrast, without shared randomness, it is difficult for the first $k - 1$ players to agree on a valid index of s . Even with quantum resources, they are forced to transmit a substantial amount of information to guarantee that the referee can find an intersection with the selector string.

We summarize our main contribution in the following informal statement that will be proved after we establish our technical tools.

► **Theorem 1 (Main Result (Informal)).** *For the k -party INDEX COORDINATION problem, classical protocols with shared randomness are exponentially more powerful than quantum protocols without prior entanglement. Specifically:*

■ **Public-coin protocols** can solve the problem with logarithmic cost:

$$R_{\text{pub}}^{\parallel}(IC_{k,n}) = O(k \log n).$$

■ **Quantum protocols** require polynomial cost:

$$Q^{\parallel}(IC_{k,n}) = \Omega\left(\frac{1}{k} n^{1-1/k}\right).$$

Crucially, this quantum lower bound becomes stronger as the number of players increases. For $k = 2$ we retrieve the bounds shown in [11] for the two-party relation.

Proof Sketch of the lower bound. Assume there exists a quantum protocol that solves $\text{IC}_{k,n}$ using messages of size m . We derive the lower bound by analyzing the limits of information transfer without shared randomness.

First, using Holevo's bound, we establish that the limited message size constrains the local success probability of identifying any specific input bit; for most indices, this probability is proportional to m/n . We then construct a hard input distribution by fixing the selector string s to be supported exclusively on these low-information indices.

To analyze the referee's performance, we apply our Direct-Product Theorem (which is informally stated below and proved in Section 3). This theorem establishes that the probability of the referee simultaneously identifying the inputs of all players at a given index is bounded by the product of the corresponding local success probabilities.

Finally, we sum these probabilities over the valid indices of s to obtain the total success probability of the protocol, which, due to our hard distribution, is dependent on the message length m . Since the protocol is required to succeed with high probability, this sum must be close to 1, which forces the message size m to be polynomially large. $\blacktriangleleft$

Generalized Zero-Error Direct-Product Theorem

The technical core of our lower bound is a new quantum complexity tool of independent interest: a direct-product theorem for zero-error state identification. Direct-product theorems generally assert that solving n instances of a problem requires $\Omega(n)$ times the resources for a single instance, even to achieve success probability of $2^{-\Omega(n)}$. While such results are established for the two-party bounded-error setting [12, 21], deriving them for general mixed states in the k -party zero-error regime presents significant challenges. The main difficulty lies in characterizing the optimal measurement strategy for the zero-error regime, where the referee is constrained to maintain perfect correctness across a tensor product of k independent quantum states.

In the two-party analysis of Gavinsky et al. [11], the lower bound relies on a straightforward factorization of the quantum state supports. However, such factorization is highly non-trivial for the arbitrary k -register case, where the intersection of multiple subspaces complicates the analysis of zero-error measurements. We overcome this obstacle by proving that a clean factorization is indeed possible. We summarize this result below:

► **Theorem 2 (Direct-Product Theorem (Informal)).** *Consider a referee receiving a product state $\rho = \bigotimes_{j=1}^k \rho_j$, where each ρ_j encodes a part of a composite label $b \in \{0,1\}^k$. The probability p_{global} of unambiguously identifying the full label b is bounded by the product of the optimal local success probabilities p_j :*

$$p_{\text{global}} \leq 2^k \prod_{j=1}^k p_j.$$

Proof Sketch. The proof relies on decomposing the global Hilbert space under zero-error constraints. To identify a specific composite label $b = (b_1, \dots, b_k)$ without error, the referee is restricted to measuring within a specific subspace $\mathcal{I}$ that is orthogonal to every incorrect string b' . While this subspace is defined globally as the intersection of kernels of all incorrect states, we prove a structural lemma showing that it factorizes cleanly into a tensor product

of local subspaces, $\mathcal{I} \cong \mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_k$, where each $\mathcal{S}_j$ is the subspace that optimally identifies the bit b_j locally.

This factorization implies that the optimal global measurement decomposes into independent local measurements. Therefore, the probability of identifying the global state ρ_b becomes the product of the probabilities of identifying each local component. Bounding these individual terms by the optimal local success probabilities a_j yields the final multiplicative bound $p \leq 2^k \prod a_j$. ◀

This multiplicative bound is the mechanism that allows us to lift local information bounds to the global multiparty setting. It ensures that the difficulty of the coordination task scales with the number of players, driving the exponential separation observed in our main result.

1.2 Related Work

The study of quantum versus classical power in the SMP model began with the quantum fingerprinting technique of Buhrman et al. [5], which showed that quantum protocols can outperform private-coin protocols.

The question of whether quantum protocols could match public coins was addressed in the one-way communication model by Bar-Yossef et al. [4] via the HIDDEN MATCHING problem, and later in the SMP model by Gavinsky et al. [13], demonstrating exponential gaps in favor of quantum protocols over classical ones.

The converse direction, where public coins outperform quantum messages, was established for two players by Gavinsky et al. [12]. Our work can be viewed as the multiparty, zero-error generalization of this result.

In the context of multiparty lower bounds, significant effort has been directed toward the Number-on-the-Forehead (NOF) model. Kerenidis [15] and Lee et al. [18] developed powerful frameworks for this setting. However, proving large quantum–classical gaps in the NOF model is notoriously difficult. Lee and Shraibman [18] (and follow-up work) showed that, for total functions, the approximation norms that *lower bound* randomized and quantum NOF communication are polynomially related. Consequently, norm-based lower bound techniques alone cannot yield exponential separations between quantum and classical complexity in this model.

Very recently, progress has been made using “lifting theorems” to transfer two-party hardness to the multiparty setting. Yang and Zhang [23] established deterministic lifting theorems for NOF, and [24] reported separations in the 3-player NOF model. Furthermore, in recent years the relation between public randomness and quantum resources has been a subject of major interest in the context of distributed computing [7, 8], with developments in the CONGEST model [9, 10], and surprising separations for locally checkable problems in the LOCAL model [2, 3].

2 Preliminaries

In this section, we establish the notation and fundamental concepts required for our results, including the formal definition of the SMP model and the basics of zero-error state discrimination.

2.1 Notation and Basic Concepts

We assume familiarity with standard quantum information notation [16, 19]. All Hilbert spaces are finite-dimensional and denoted by $\mathcal{H}$. We write $L(\mathcal{H})$ for the set of linear operators

on $\mathcal{H}$ and

$$\mathcal{D}(\mathcal{H}) := \{\rho \in \mathcal{L}(\mathcal{H}) : \rho \succeq 0, \text{Tr}(\rho) = 1\}$$

for the set of density operators. Here $\rho \succeq 0$ means that ρ is positive semidefinite, i.e., $\langle v, \rho v \rangle \geq 0$ for all $v \in \mathcal{H}$, and more generally $A \succeq B$ abbreviates $A - B \succeq 0$.

For an operator $A \in \mathcal{L}(\mathcal{H})$, the trace is

$$\text{Tr}(A) = \sum_i \langle e_i, A e_i \rangle,$$

where $\{e_i\}$ is any orthonormal basis of $\mathcal{H}$. The support of an operator ρ is denoted $\text{supp}(\rho)$, and its kernel by $\ker(\rho)$. If $S \subseteq \mathcal{H}$ is a subspace, then $S^\perp$ denotes its orthogonal complement and Π_S the orthogonal projector onto S . Composite systems are written as tensor products, e.g., $\mathcal{H} = \bigotimes_{j=1}^k \mathcal{H}_j$, and we use subscripts such as ρ_{AB} to indicate the registers on which a state acts.

The von Neumann entropy of a state ρ is

$$S(\rho) = -\text{Tr}(\rho \log \rho),$$

and the mutual information between two registers X and Y is

$$I(X : Y) = S(X) + S(Y) - S(XY).$$

We freely identify classical random variables with quantum systems in diagonal states: if X takes values in a finite set $\mathcal{X}$ with distribution $(p_x)_{x \in \mathcal{X}}$, we represent it by the density matrix

$$\rho_X = \sum_{x \in \mathcal{X}} p_x |x\rangle\langle x|,$$

for some fixed orthonormal basis $\{|x\rangle : x \in \mathcal{X}\}$. With this identification we use the same notation for classical and quantum entropies and mutual information. We use $[n]$ to denote the set $\{1, \dots, n\}$.

2.2 Communication Model and the Zero-Error Condition

The Simultaneous Message Passing (SMP) model involves k spatially separated players $P_1, \dots, P_k$ and a central referee R . Each player P_j receives a private input x_j from a domain X_j . The players cannot communicate with each other; instead, they each send a single message to the referee. All parties know in advance a relational problem

$$R \subseteq X_1 \times \dots \times X_k \times Z,$$

and their joint goal, on input $(x_1, \dots, x_k)$, is to produce some output $z \in Z$ such that $(x_1, \dots, x_k, z) \in R$.

We analyze this model in the zero-error (unambiguous) regime. In this setting, the referee is permitted to output a special failure symbol $\perp$, but is strictly forbidden from producing an incorrect answer. Formally, a protocol computes R with error parameter $\varepsilon \in [0, 1)$ if, for all inputs $(x_1, \dots, x_k)$:

- *Unambiguous output:* Whenever the referee outputs some $z \neq \perp$, we have $(x_1, \dots, x_k, z) \in R$ with probability 1.
- *Success probability:* The probability that the referee outputs $\perp$ is at most ε .

The communication cost of a protocol is the worst-case length of the longest message sent by any single player, taken over all inputs and all internal randomness. We compare this cost across three resource models:

- **Private-coin** ($R_{\text{priv}}^{\parallel}$): players have access only to private sources of randomness, and the referee sees only the resulting messages.
- **Public-coin** ($R_{\text{pub}}^{\parallel}$): all players (and the referee) share a common random string, which does not count toward the communication cost.
- **Quantum** ($Q^{\parallel}$): each player P_j sends a quantum state $\rho_j(x_j)$ as a message; the referee may apply any quantum operation and POVM on the joint state $\bigotimes_{j=1}^k \rho_j(x_j)$. The players do not share any prior entanglement.

2.3 Unambiguous State Discrimination

A Positive Operator-Valued Measure (POVM) $\{E_0, E_1, E_{\perp}\}$ unambiguously distinguishes a pair (α_0, α_1) if $\text{Tr}(E_0 \alpha_1) = 0$ and $\text{Tr}(E_1 \alpha_0) = 0$. The optimal one-sided probability of identifying α_0 is $a^{(0)} := \text{Tr}(\alpha_0 \Pi_{\ker \alpha_1})$. See [6, 20] for further details on this topic.

We rely on two structural properties for the analysis of product states. First, zero-error discrimination is non-trivial only within the combined support.

► **Proposition 3** (Restriction to Sum of Supports [20]). *Let $\alpha_0, \alpha_1 \in \mathcal{D}(\mathcal{H})$. Define the subspace $S := \text{supp}(\alpha_0) + \text{supp}(\alpha_1) \subseteq \mathcal{H}$. Any positive operator E satisfying the zero-error constraint $\text{Tr}(E \alpha_i) = 0$ (for $i \in \{0, 1\}$) must satisfy $\text{supp}(E) \subseteq (\text{supp}(\alpha_i))^{\perp}$.*

Consequently, the unambiguous discrimination of α_0 and α_1 on $\mathcal{H}$ is equivalent to discriminating their restrictions to the subspace $S := \text{supp}(\alpha_0) + \text{supp}(\alpha_1)$.

The second property is that trace of tensor products factorizes, facilitating component-wise analysis.

► **Proposition 4** (Trace Factorization [19, 22]). *Let $\mathcal{H}, \mathcal{K}$ be finite-dimensional Hilbert spaces. For operators $A, C \in \mathcal{L}(\mathcal{H})$ and $B, D \in \mathcal{L}(\mathcal{K})$, $\text{Tr}((A \otimes B)(C \otimes D)) = \text{Tr}(AC) \cdot \text{Tr}(BD)$.*

2.4 Information Theoretic Tools

To lower bound communication, we bound the accessible classical information using Holevo's theorem.

► **Proposition 5** (Holevo's Bound [14, 19]). *Let X be a random variable over $\{1, \dots, n\}$ with distribution p_x , encoded into quantum states ρ_x . If a measurement on ρ_X yields a classical outcome Y , then the mutual information satisfies $I(X : Y) \leq S(\sum_x p_x \rho_x) - \sum_x p_x S(\rho_x)$.*

In particular, since $S(\sum_x p_x \rho_x) \leq \log_2 \dim \mathcal{H}$ we have

$$I(X : Y) \leq \log_2 \dim \mathcal{H}.$$

This bound establishes that the amount of classical information that can be extracted from an m -qubit state is m bits.

3 Direct-Product Theorem for Zero-Error Identification

In the multiparty SMP model, the referee receives a product state $\rho = \bigotimes_{j=1}^k \rho_j$. To solve certain relational problems involving the players' input bits, the referee must identify which label $b \in \{0, 1\}^k$ (corresponding to the players' distributed inputs) is encoded in this product

state. Our goal in this section is to upper bound the best possible success probability of such zero-error identification tasks.

Informally, we prove that in the zero-error regime the *global* success probability of identifying the full label b is at most the product of the corresponding *local* success probabilities for the individual registers, up to a factor 2^k ; see Theorem 1 below. Direct-product theorems of this form are well known in two-party communication complexity, where they typically state that solving many independent instances of a task requires resources that scale essentially linearly in the number of instances. Here the task is unambiguous identification of a label from a tensor product of mixed states, and the challenge is to characterize the optimal measurement strategy under the zero-error constraint.

For $k = 2$, Gavinsky et al. [11] obtained a lower bound by exploiting a simple factorization of the supports of the relevant quantum states. For general k , such a factorization is far from obvious: the intersection of the many local subspaces leads to a complicated global measurement subspace. Our main technical contribution is to show that, despite this complexity, the subspace relevant for zero-error identification still factorizes cleanly as a tensor product of local subspaces. This structural statement is captured in Lemma 1 below and is the key ingredient in the proof of Theorem 1, which we now restate in a formal way.

► **Theorem 1 (Zero-Error Product Identification).** *Let $\{(\alpha_j^{(0)}, \alpha_j^{(1)})\}_{j=1}^k$ be pairs of mixed states. We define the single-register zero-error success probability as*

$$a_j := \sup_M \frac{1}{2} \left(\text{Tr}(E_0 \alpha_j^{(0)}) + \text{Tr}(E_1 \alpha_j^{(1)}) \right),$$

where the supremum is taken over all zero-error POVMs $M = \{E_0, E_1, E_\perp\}$ on $\mathcal{H}_j$.

Consider the ensemble of 2^k product states

$$\rho_b = \bigotimes_{j=1}^k \alpha_j^{(b_j)}, \quad b \in \{0, 1\}^k,$$

where each label b is chosen with equal probability 2^{-k} . Let p denote the maximal success probability of a zero-error measurement that distinguishes this ensemble under the uniform prior, i.e., the quantity $p = 2^{-k} \sum_{b \in \{0, 1\}^k} p_b$, where p_b is the conditional probability of correctly outputting b when ρ_b is sent. It follows that

$$p \leq 2^{-k} \prod_{j=1}^k a_j.$$

This generalization is crucial for deriving the communication lower bounds in the subsequent sections. To prove the theorem, we require a structural result characterizing the intersection of orthogonal complements in tensor product spaces.

One key difficulty in analyzing product states is determining the optimal measurement subspace [11, 12, 20]. The following algebraic lemma characterizes the intersection of the orthogonal complements of tensor product spaces.

► **Lemma 1 (Subspace-Complement Factorization).** *Let $k \in \mathbb{N}$. For each $j \in [k]$, let $\mathcal{H}_j$ be a Hilbert space composed of subspaces $S_j^{(0)}, S_j^{(1)} \subseteq \mathcal{H}_j$ satisfying the condition $\mathcal{H}_j = S_j^{(0)} + S_j^{(1)}$. Fix a binary string $b' \in \{0, 1\}^k$. We define two subspaces of the tensor-product space $\mathcal{H} = \bigotimes_{j=1}^k \mathcal{H}_j$ as*

$$X = \bigcap_{b \in \{0, 1\}^k \setminus \{b'\}} \left(\bigotimes_{j=1}^k S_j^{(b_j)} \right)^\perp, \quad Y = \bigotimes_{j=1}^k (S_j^{(1 \oplus b'_j)})^\perp.$$

It follows that $X = Y$.

Proofs for Lemma 1 and Theorem 1 can be found in Appendix A

4 Exponential Separation in Multipart Communication

In this section, we present our main result: an exponential separation between the public-coin and quantum SMP models for a multipart relational problem. We introduce the problem INDEX COORDINATION ($\text{IC}_{k,n}$) and demonstrate that while it admits an efficient logarithmic-cost protocol using public randomness, any zero-error quantum protocol (without shared entanglement) requires polynomial communication.

- **Definition 6** (Problem $\text{IC}_{k,n}$). *The problem involves k players $P_1, \dots, P_k$.*
- **Input:** *Player $P_j \in \{1, \dots, k-1\}$ receives a string $x_j \in \{0, 1\}^n$. Player P_k receives a string $x_k \in \{0, 1\}^n$ and a selector string $s \in \{0, 1\}^n$ with Hamming weight $|s| = n/k$, i.e., exactly n/k positions are set to 1.*
 - **Output:** *The referee must output a tuple $(i, x_1^{(i)}, \dots, x_k^{(i)})$ for some index i such that $s^{(i)} = 1$.*
 - **Constraint:** *The protocol must be zero-error. The referee may output $\perp$ with probability at most ε , but any non- $\perp$ output must be correct.*

4.1 Classical Protocols

Public-Coin Upper Bound.

This problem admits a straightforward classical protocol with shared randomness. The players use the public coin to sample indices $i \in [n]$ uniformly at random. Since the target set (indices where $s^{(i)} = 1$) has relative frequency $1/k$, a random index is valid with probability $1/k$. Upon selecting an index, all players send their corresponding bit $x_j^{(i)}$ and index i , player P_k also sends the bit $s^{(i)}$, confirming whether $s^{(i)} = 1$. Repeating this process $O(k \log(1/\varepsilon))$ times ensures a successful output with high probability. Hence

$$R_{\text{pub}}^{\parallel}(\text{IC}_{k,n}) = O(k \log n \cdot \log(1/\varepsilon)).$$

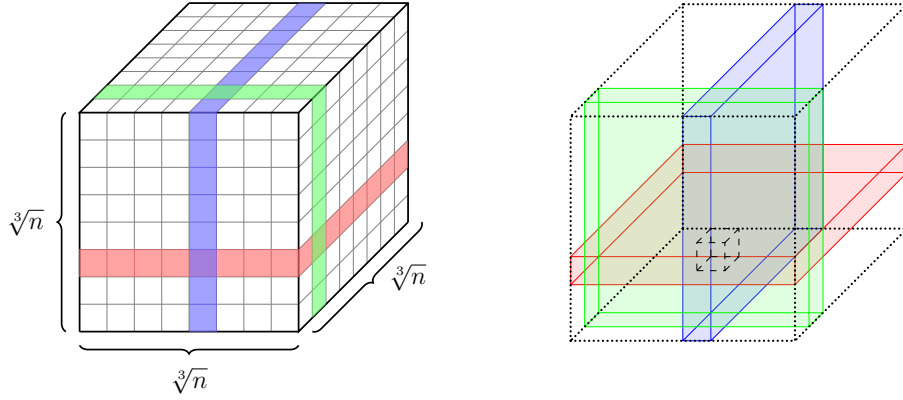
Private-Coin Baseline.

Without public randomness, players cannot coordinate on a specific index efficiently. The best known strategy is a generalization of the protocol described in [1]. Each player treats their n -bit input as a flattened k -rank tensor of dimension $n^{1/k} \times \dots \times n^{1/k}$. Each player P_j selects a random coordinate $c_j \in [n^{1/k}]$ for the j -th dimension and sends the corresponding sub-tensor (of rank $k-1$) obtained by fixing that coordinate. Geometrically, these sub-tensors are hyperplanes orthogonal to different axes (see Figure 1). Their intersection is exactly one index i . The referee checks if this intersection index i satisfies $s_i = 1$. Figure 1 shows a geometric intuition for this protocol.

Communication cost is then proportional to the size of the sub-tensor sent by players:

$$R_{\text{priv}}^{\parallel}(\text{IC}_{k,n}) = O\left(k \cdot n^{1-1/k}\right).$$

This upper bound provides a baseline: Private randomness can solve the coordination problem using polynomial bits. Since quantum protocols can simulate private randomness, this also establishes an upper bound for the quantum model.



(a) Each color represents an index set chosen by one player, in this case each player chooses a matrix (2-rank tensor) from the construction

(b) Three orthogonal matrices always intersect in a common index, this is the referee selection

■ Figure 1

4.2 Quantum Lower Bound

We now prove that quantum messages offer no asymptotic advantage over private coins for this problem, establishing an exponential gap compared to the public-coin model.

► **Theorem 7.** *Any zero-error quantum SMP protocol for $\text{IC}_{k,n}$ without prior entanglement requires a message size of $m = \Omega\left(\frac{1}{k}n^{1-1/k}\right)$ qubits per player.*

Proof. Let m be the message size in qubits. We denote the message of player P_j on input x_j as $\rho_j(x_j)$. To bound m , we analyze the information capacity of these messages regarding specific input bits and apply our Direct-Product Theorem (Theorem 1).

Step 1: Identifying high-information indices.

Fix a player $j \in [k-1]$ and an index $i \in [n]$. Consider the two mixed states obtained by averaging the message of P_j over all inputs with a fixed value of the i -th bit:

$$\rho_j^{(i,0)} := \mathbb{E}_{x_j: x_j^{(i)}=0} [\rho_j(x_j)], \quad \rho_j^{(i,1)} := \mathbb{E}_{x_j: x_j^{(i)}=1} [\rho_j(x_j)].$$

We then define the single-bit zero-error success probability

$$a_j^{(i)} := \sup_M \frac{1}{2} \left(\text{Tr}(E_0 \rho_j^{(i,0)}) + \text{Tr}(E_1 \rho_j^{(i,1)}) \right),$$

where the supremum is taken over all zero-error POVMs $M = \{E_0, E_1, E_\perp\}$ acting on the message space of P_j (i.e., POVMs satisfying $\text{Tr}(E_0 \rho_j^{(i,1)}) = \text{Tr}(E_1 \rho_j^{(i,0)}) = 0$). Intuitively, $a_j^{(i)}$ is the optimal probability with which the referee can learn the i -th bit of x_j from the quantum message alone, while never making an error.

Applying the Holevo bound (Proposition 5), the m -qubit average message cannot carry more than m bits of information. Hence

$$\sum_{i=1}^n a_j^{(i)} \leq m.$$

We identify the “bad” indices where a player’s message reveals too much information. Applying Markov’s inequality with a threshold of km/n we have

$$\Pr_{i \in [n]} \left[a_j^{(i)} \geq \frac{km}{n} \right] \leq \frac{\mathbb{E}[a_j^{(i)}]}{km/n} \leq \frac{m/n}{km/n} = \frac{1}{k}.$$

Let $B_j = \{i \in [n] : a_j^{(i)} > \frac{km}{n}\}$ be the set of high-information indices for player P_j . The bound implies $|B_j| \leq n/k$.

Step 2: Constructing a hard instance.

We construct a hard input distribution by excluding these high-information sets. Consider the union of the “bad” sets across all players $j = 1, \dots, k-1$. The size of this union is at most

$$\left| \bigcup_{j=1}^{k-1} B_j \right| \leq \sum_{j=1}^{k-1} |B_j| \leq (k-1) \frac{n}{k} = n - \frac{n}{k}.$$

Let $G = [n] \setminus \bigcup_{j=1}^{k-1} B_j$ be the set of “good” indices. The size of this set is at least

$$|G| \geq n - \left(n - \frac{n}{k} \right) = \frac{n}{k}.$$

For every index $i \in G$, every player $j \in [k-1]$ satisfies the tight information bound $a_j^{(i)} \leq \frac{km}{n}$. The hard instance we consider is then produced by fixing player P_k ’s input selector s to be the indicator vector of n/k distinct indices chosen exclusively from G .

Step 3: Bounding the referee’s success probability.

The referee receives the product state $\bigotimes_{j=1}^k \rho_j$. To succeed, the referee must identify the tuple $(x_1^{(i)}, \dots, x_k^{(i)})$ for some i where $s^{(i)} = 1$ (which implies $i \in G$). Let p_i be the probability that the referee correctly outputs a valid tuple for index i . By the Zero-Error Product Identification Theorem (Theorem 1), we have

$$p_i \leq 2^k \prod_{j=1}^k a_j^{(i)} = 2^k a_k^{(i)} \prod_{j=1}^{k-1} a_j^{(i)}.$$

Since $i \in G$, we substitute the bound $a_j^{(i)} \leq \frac{km}{n}$ for $j < k$,

$$p_i \leq 2^k a_k^{(i)} \left(\frac{km}{n} \right)^{k-1}.$$

The total success probability of the protocol is then bounded by the sum over all valid indices

$$\begin{aligned} 1 - \varepsilon &\leq \sum_{i \in G} p_i \\ &\leq \sum_{i \in G} 2^k a_k^{(i)} \prod_{j=1}^{k-1} a_j^{(i)} && \text{(Theorem 1)} \\ &\leq 2^k \sum_{i \in G} a_k^{(i)} \left(\frac{km}{n} \right)^{k-1} && \text{(Since } i \in G) \end{aligned}$$

$$= 2^k \left(\frac{km}{n} \right)^{k-1} \sum_{i \in G} a_k^{(i)}.$$

Using the bound for player P_k , $\sum_{i \in G} a_k^{(i)} \leq \sum_{i=1}^n a_k^{(i)} \leq m$. Thus

$$1 - \varepsilon \leq 2^k k^{k-1} \frac{m^{k-1}}{n^{k-1}} \cdot m = \frac{2^k k^{k-1} m^k}{n^{k-1}}.$$

Rearranging:

$$m = \Omega \left(\frac{1}{k} \cdot n^{(k-1)/k} \right) = \Omega \left(\frac{1}{k} \cdot n^{1-1/k} \right).$$

◀

4.3 Main Result

Combining the bounds derived above, we establish the following exponential separation for the problem $\text{IC}_{k,n}$, showing quantum messages offer no asymptotic advantage over private randomness, and both are exponentially weaker than public randomness.

► **Theorem 8** (Exponential Separation). *For the multiparty relational problem $\text{IC}_{k,n}$, the communication complexities in the public-coin, private-coin, and quantum SMP models satisfy*

$$\begin{aligned} R_{\text{pub}}^{\parallel}(\text{IC}_{k,n}) &= O(k \log n), & R_{\text{priv}}^{\parallel}(\text{IC}_{k,n}) &= O(k \cdot n^{1-1/k}), \\ Q^{\parallel}(\text{IC}_{k,n}) &= \Omega \left(\frac{1}{k} \cdot n^{1-1/k} \right). \end{aligned}$$

When k is a constant, we observe $Q^{\parallel}(\text{IC}_{k,n}) = \Theta(n^{1-1/k})$ versus $R_{\text{pub}}^{\parallel}(\text{IC}_{k,n}) = O(\log n)$.

Conclusion

We have presented a direct-product theorem for the zero-error identification of mixed product states and applied it to the multiparty SMP model. Our results establish that for the relational problem $\text{IC}_{k,n}$, public randomness is exponentially more powerful than quantum communication without prior entanglement.

This work addresses the gap in zero-error composition theorems for mixed states, providing a tool that may find further applications in distributed complexity and quantum information theory.

An open question is whether these separations extend beyond relational problems to functional (Boolean) ones in the multiparty zero-error or in the even more general bounded-error settings.

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A Omitted Proofs

We present the proofs of technical Lemma 1 and Theorem 1, restated for clarity.

► **Lemma 1** (Subspace-Complement Factorization). *Let $k \in \mathbb{N}$. For each $j \in [k]$, let $\mathcal{H}_j$ be a Hilbert space composed of subspaces $S_j^{(0)}, S_j^{(1)} \subseteq \mathcal{H}_j$ satisfying the condition $\mathcal{H}_j = S_j^{(0)} + S_j^{(1)}$. Fix a binary string $b' \in \{0, 1\}^k$. We define two subspaces of the tensor-product space $\mathcal{H} = \bigotimes_{j=1}^k \mathcal{H}_j$ as*

$$X = \bigcap_{b \in \{0,1\}^k \setminus \{b'\}} \left(\bigotimes_{j=1}^k S_j^{(b_j)} \right)^\perp, \quad Y = \left(\bigotimes_{j=1}^k (S_j^{(1 \oplus b'_j)})^\perp \right)^\perp.$$

It follows that $X = Y$.

Proof. Since X and Y are closed linear subspaces of $\mathcal{H}$, and the set of decomposable (pure) tensors spans $\mathcal{H}$, it suffices to verify equality for pure tensors. Indeed, if $\{|e_j^{(\ell)}\rangle : \ell \in I_j\}$ is an orthonormal basis of $\mathcal{H}_j$ for each $j \in [k]$, then the elementary tensors

$$\{|e_1^{(\ell_1)}\rangle \otimes \cdots \otimes |e_k^{(\ell_k)}\rangle : (\ell_1, \dots, \ell_k) \in I_1 \times \cdots \times I_k\}$$

form an orthonormal basis of $\mathcal{H}$. Every vector in $\mathcal{H}$ is therefore a linear combination of decomposable pure tensors. We establish the result via double inclusion.

(i) **Inclusion $X \subseteq Y$.** Let $|v\rangle = \bigotimes_{j=1}^k |v_j\rangle$ be a non-zero pure tensor belonging to X . To show it belongs to Y , we must show that for every index $t \in [k]$, the factor $|v_t\rangle$ resides in the subspace $(S_t^{(1 \oplus b'_t)})^\perp$. We proceed by contradiction. Assume there exists an index $t \in [k]$ such that $|v_t\rangle \notin (S_t^{(1 \oplus b'_t)})^\perp$. We construct a specific binary string $b \in \{0, 1\}^k$ based on the properties of the factors of $|v\rangle$. For each $j \in [k]$, let

$$b_j = \begin{cases} 1 \oplus b'_j & \text{if } |v_j\rangle \notin (S_j^{(1 \oplus b'_j)})^\perp, \\ b'_j & \text{otherwise.} \end{cases}$$

By the assumption on index t , we have $b_t = 1 \oplus b'_t$, which implies that $b \neq b'$. We now assert that for every $j \in [k]$, there exists a state $|s_j\rangle \in S_j^{(b_j)}$ such that the inner product $\langle v_j | s_j \rangle$ is non-zero. We analyze the two cases for b_j .

- **Case 1:** $b_j = 1 \oplus b'_j$. By the definition of b , this implies $|v_j\rangle \notin (S_j^{(b_j)})^\perp$. Consequently, there must exist some $|s_j\rangle \in S_j^{(b_j)}$ that is not orthogonal to $|v_j\rangle$.

- **Case 2:** $b_j = b'_j$. In this scenario, the construction of b implies $|v_j\rangle \in (S_j^{(1 \oplus b'_j)})^\perp$. Recall that $\mathcal{H}_j = S_j^{(0)} + S_j^{(1)}$. Taking orthogonal complements yields $\{0\} = \mathcal{H}_j^\perp = (S_j^{(0)})^\perp \cap (S_j^{(1)})^\perp$. Since $|v_j\rangle$ is a non-zero vector orthogonal to $S_j^{(1 \oplus b'_j)}$, it cannot also be orthogonal to $S_j^{(b'_j)}$. Thus, $|v_j\rangle$ must have a non-zero overlap with $S_j^{(b'_j)} = S_j^{(b_j)}$, ensuring the existence of the required state $|s_j\rangle$.

Define the product state $|s\rangle = \bigotimes_{j=1}^k |s_j\rangle$. By construction, $|s\rangle \in \bigotimes_{j=1}^k S_j^{(b_j)}$. Calculating the inner product yields

$$\langle v|s\rangle = \prod_{j=1}^k \langle v_j|s_j\rangle \neq 0.$$

This result implies that $|v\rangle$ is not orthogonal to the subspace $\bigotimes_{j=1}^k S_j^{(b_j)}$. However, since $|v\rangle \in X$ and $b \neq b'$, the definition of X requires that $|v\rangle \perp \bigotimes_{j=1}^k S_j^{(b_j)}$. We have a contradiction. Therefore, for all t , $|v_t\rangle \in (S_t^{(1 \oplus b'_t)})^\perp$. We conclude that $|v\rangle \in Y$.

(ii) **Inclusion** $Y \subseteq X$. Conversely, let $|v\rangle = \bigotimes_{j=1}^k |v_j\rangle$ be an element of Y . By the definition of Y , for every $j \in [k]$, we have $|v_j\rangle \in (S_j^{(1 \oplus b'_j)})^\perp$. To verify that $|v\rangle \in X$, consider an arbitrary binary string $b \in \{0, 1\}^k$ such that $b \neq b'$. The inequality $b \neq b'$ implies the existence of at least one index t where $b_t \neq b'_t$, or equivalently, $b_t = 1 \oplus b'_t$. Consequently, the hypothesis on $|v\rangle$ implies

$$|v_t\rangle \in (S_t^{(1 \oplus b'_t)})^\perp = (S_t^{(b_t)})^\perp.$$

It follows that $\langle v_t|s_t\rangle = 0$ for any vector $|s_t\rangle \in S_t^{(b_t)}$. Now, consider any product state $|s\rangle = \bigotimes_{j=1}^k |s_j\rangle$ within the subspace $\bigotimes_{j=1}^k S_j^{(b_j)}$. The total inner product decomposes as $\langle v|s\rangle = \prod_{j=1}^k \langle v_j|s_j\rangle$. Since the term corresponding to index t vanishes, the entire product is zero, meaning $|v\rangle \perp |s\rangle$. Extending this by linearity, $|v\rangle$ is orthogonal to the entire subspace $\bigotimes_{j=1}^k S_j^{(b_j)}$. As b was an arbitrary string distinct from b' , this holds for all such strings, thereby proving $|v\rangle \in X$. ◀

► **Theorem 1 (Zero-Error Product Identification).** Let $\{(\alpha_j^{(0)}, \alpha_j^{(1)})\}_{j=1}^k$ be pairs of mixed states. We define the single-register zero-error success probability as

$$a_j := \sup_M \frac{1}{2} \left(\text{Tr}(E_0 \alpha_j^{(0)}) + \text{Tr}(E_1 \alpha_j^{(1)}) \right),$$

where the supremum is taken over all zero-error POVMs $M = \{E_0, E_1, E_\perp\}$ on $\mathcal{H}_j$.

Consider the ensemble of 2^k product states

$$\rho_b = \bigotimes_{j=1}^k \alpha_j^{(b_j)}, \quad b \in \{0, 1\}^k,$$

where each label b is chosen with equal probability 2^{-k} . Let p denote the maximal success probability of a zero-error measurement that distinguishes this ensemble under the uniform prior, i.e., the quantity $p = 2^{-k} \sum_{b \in \{0, 1\}^k} p_b$, where p_b is the conditional probability of correctly outputting b when ρ_b is sent. It follows that

$$p \leq 2^{-k} \prod_{j=1}^k a_j.$$

Proof of Theorem 1. Fix a register j . For the pair of states $(\alpha_j^{(0)}, \alpha_j^{(1)})$, define the one-sided zero-error success probabilities

$$a_j^{(0)} := \sup_M \text{Tr}(E_0 \alpha_j^{(0)}), \quad a_j^{(1)} := \sup_M \text{Tr}(E_1 \alpha_j^{(1)}),$$

where the supremum is taken over all zero-error POVMs $M = \{E_0, E_1, E_\perp\}$ on $\mathcal{H}_j$. We recall the definition of single-register zero-error success probability

$$a_j := \sup_M \frac{1}{2} \left(\text{Tr}(E_0 \alpha_j^{(0)}) + \text{Tr}(E_1 \alpha_j^{(1)}) \right),$$

Note that

$$a_j^{(0)}, a_j^{(1)} \leq 2a_j. \quad (1)$$

Indeed, if one applies a zero-error measurement that optimally identifies, say, $\alpha_j^{(0)}$, then the average success probability is at least $\frac{1}{2}a_j^{(0)}$, and by maximality of a_j this implies $a_j \geq \frac{1}{2}a_j^{(0)}$. The same argument applies to $a_j^{(1)}$.

Let $p^{(b)}$ denote the maximal conditional probability of correctly identifying the state ρ_b with zero error. Since the prior over labels is uniform, we have

$$p \leq \frac{1}{2^k} \sum_{b \in \{0,1\}^k} p^{(b)}. \quad (2)$$

We now compute $p^{(b)}$. Fix $b = 0^k$; the argument for general b is identical. Let $\mathcal{S}_j^{(0)}$ and $\mathcal{S}_j^{(1)}$ denote the supports of $\alpha_j^{(0)}$ and $\alpha_j^{(1)}$, respectively. By Proposition 3, we may assume $\mathcal{H}_j = \mathcal{S}_j^{(0)} + \mathcal{S}_j^{(1)}$.

The optimal one-sided success probability for $\alpha_j^{(0)}$ is achieved by projecting onto the orthogonal complement of the support of $\alpha_j^{(1)}$

$$a_j^{(0)} = \text{Tr}(\alpha_j^{(0)} \Pi_{(\mathcal{S}_j^{(1)})^\perp}).$$

To identify $\rho_{0^k} = \bigotimes_{j=1}^k \alpha_j^{(0)}$ with zero error, any measurement operator Π that achieves this identification must have support orthogonal to every other product state $\rho_{b'}$ with $b' \neq 0^k$. Equivalently, the support of Π is contained in the intersection

$$\mathcal{I} = \bigcap_{b' \neq 0^k} \ker(\rho_{b'}) = \bigcap_{b' \neq 0^k} \left(\bigotimes_{j=1}^k \mathcal{S}_j^{(b'_j)} \right)^\perp.$$

By Lemma 1, this intersection factorizes, so that

$$\mathcal{I} = \bigotimes_{j=1}^k (\mathcal{S}_j^{(1)})^\perp.$$

Therefore the orthogonal projector onto $\mathcal{I}$ decomposes as the tensor product of the local orthogonal projectors, $\Pi_{\mathcal{I}} = \bigotimes_{j=1}^k \Pi_{(\mathcal{S}_j^{(1)})^\perp}$, and under the zero error constraint this projector maximizes the detection probability for ρ_{0^k} . Consequently, the success probability for ρ_{0^k} equals

$$p^{(0^k)} = \text{Tr} \left[\left(\bigotimes_{j=1}^k \alpha_j^{(0)} \right) \Pi_{\mathcal{I}} \right].$$

Applying Proposition 4 to factorize the trace of a tensor product against a tensor product projector yields

$$p^{(0^k)} = \prod_{j=1}^k \text{Tr}(\alpha_j^{(0)} \Pi_{(\mathcal{S}_j^{(1)})^\perp}) = \prod_{j=1}^k a_j^{(0)}.$$

The same argument applied to an arbitrary binary string b gives the analogous identity

$$p^{(b)} = \prod_{j=1}^k a_j^{(b_j)}.$$

Substituting these expressions into Equation (2) and invoking the componentwise bound from Equation (1), which asserts $a_j^{(b_j)} \leq 2a_j$ for each j and each bit b_j , we obtain

$$\begin{aligned} p &\leq \frac{1}{2^k} \sum_{b \in \{0,1\}^k} \prod_{j=1}^k a_j^{(b_j)} \\ &\leq \frac{1}{2^k} \sum_{b \in \{0,1\}^k} \prod_{j=1}^k (2a_j) = \frac{1}{2^k} \cdot 2^k \prod_{j=1}^k (2a_j) = \prod_{j=1}^k (2a_j) \\ &= 2^k \prod_{j=1}^k a_j. \end{aligned}$$

This concludes the proof. ◀

B Asymptotic Analysis of the Separation

In the previous section, we established an exponential separation for problem $\text{IC}_{k,n}$ when considering a fixed number of players. However, in distributed computing contexts, the number of players k often scales with the input size n . In this section, we analyze the asymptotic relationship between the input length n and the number of players k to determine the threshold where this separation breaks down.

Let $Q(n, k)$ denote the lower bound on the quantum message size derived in Section 4, and let $P(n, k)$ denote the classical public-coin upper bound. From our results

$$Q(n, k) \geq \frac{1}{2} \left(\frac{n^{k-1}}{k^{k-1}} \right)^{1/k} = \Theta \left(\frac{n^{1-1/k}}{k} \right),$$

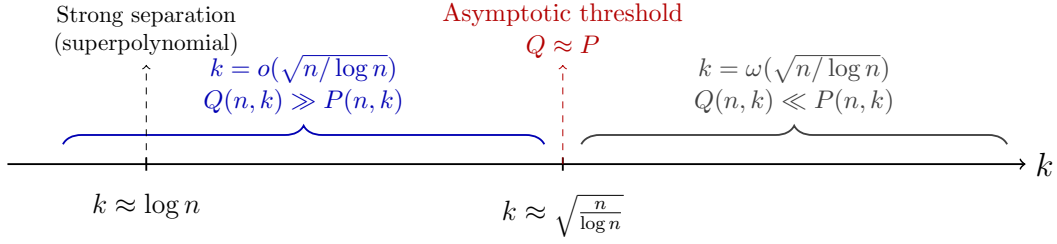
$$P(n, k) = O(k \log n).$$

We analyze the ratio of these costs in the logarithmic domain to identify the dominance regimes. Let $\Delta(n, k) = \log(Q(n, k)) - \log(P(n, k))$. Neglecting low-order terms

$$\Delta(n, k) \approx \left(1 - \frac{1}{k} \right) \log n - 2 \log k - \log \log n. \quad (3)$$

The behavior of $\Delta(n, k)$ reveals three distinct scaling regimes based on the growth of k relative to n .

► **Theorem 9 (Asymptotic Thresholds).** *For the $\text{IC}_{k,n}$ bounds previously shown, the asymptotic relationship between quantum and public-coin complexity is governed by the scaling of k .*



■ **Figure 2** Diagram showing the comparison between $Q(n, k)$ and the public-coin bound $P(n, k)$. The asymptotic threshold is shown in red.

1. **Strong Separation Regime** ($k = o(\sqrt{n/\log n})$): If the number of players grows slowly, the term $\log n$ dominates.

$$\Delta(n, k) \rightarrow +\infty \implies Q(n, k) = \omega(P(n, k)).$$

For example, if $k = \Theta(\log n)$, we obtain a polynomial separation:

$$Q(n, k) = \Theta(n/\log n) \quad \text{vs} \quad P(n, k) = \Theta(\log^2 n).$$

2. **Threshold Regime** ($k = \Theta(\sqrt{n/\log n})$): This is the critical scaling where the models become comparable. If $k \approx \sqrt{n}$, then $2 \log k \approx \log n$. The leading terms in (3) cancel, and:

$$\Delta(n, k) = O(1) \implies Q(n, k) = \Theta(P(n, k)).$$

3. **Inversion Regime** ($k = \omega(\sqrt{n/\log n})$): If the number of players is large, the overhead of identifying a specific player (k) and the cost of the public-coin index selection ($k \log n$) outweigh the hardness of the intersection problem.

$$\Delta(n, k) \rightarrow -\infty \implies Q(n, k) = o(P(n, k)).$$

This analysis demonstrates that the exponential advantage of public randomness over quantum messages is robust for small to moderate numbers of players but vanishes when the system becomes highly distributed ($k \approx \sqrt{n/\log n}$, see Figure 2).